

A Better Hand

*The PIE contrastive *-tero- and the pre-PIE hand-noun*

Paper 2 of the *der-* project. Builds on Chavis & Claude (2026a), "The Hand That Gives" (LingBuzz v26).

[Working paper v8 — for scholarly review]

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Abstract

The PIE contrastive suffix **-tero-* is, in its oldest documented function, the systematic morpheme for spatial-direction marking across the Indo-European family. Almost every PIE local particle has a **-tero-* derivative naming its spatial contrastive — Latin *citra/ultra/intra/extra/contra*, Sanskrit *ádharma-/uttara-/ápara-/ántara-*, Avestan and Old Persian *fratarā-/apatara-/upatara-*, Hittite *kattera-*, Greek *próteros/hústeros*. The Handbook of Comparative and Historical Indo-European Linguistics states explicitly that outside Greek and Indo-Iranian, **-tero-* and its bare alternant **(e)ro-* are confined to derivations from pronouns and particles; the gradational "more X" comparative function is a Greek and Indo-Iranian innovation.

This paper proposes that the original spatial-direction function of **-tero-* follows naturally from the framework of Paper 1: pre-PIE **der-* "hand, open palm" grammaticalized through the pathway HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE, and the resulting suffix was applied productively to local particles. The mainstream consensus that **-tero-* originally marked bilateral contrast — established in every standard reference (Sihler 1995; Fortson 2010) — is what the framework is trying to explain. The *ambidexter* compound makes the bilateral presupposition explicit.

The paper makes one substantive empirical contribution beyond the etymological argument. The reversal hypothesis — that bare **(e)ro-* arises by OCP-driven suffix-initial deletion before stop-final bases, rather than that **-tero-* arose by metanalysis from bare **(e)ro-* — predicts that bare forms should cluster on stop-final bases and *t*-bearing forms on bases ending in vowels, sonorants, and fricatives. Run on the inherited PIE positional adjective paradigm in Sanskrit, Avestan, Latin, and Greek, this prediction is matched in approximately 19 of 19 etymologically transparent cases. The distributional test argues against the older "irregular extension" framing; it does not by itself decide between the framework's reversal and the metanalysis hypothesis of Bauhaus (2019) and Kahl (2024). Critically, however, Bauhaus's own data inventory — the bare-suffix locatives he reconstructs in defense of metanalysis — consists almost entirely of bases ending in stops and stop-clusters, exactly the

phonological environment where the OCP-driven rule predicts the /t/ should drop. Both hypotheses fit the distributional data; the framework's advantage is etymological: it explains why the *t-* was there in the first place.

1. Introduction

The **-tero-* suffix has been thoroughly documented in the Indo-European literature, but it has never been satisfactorily explained. The standard handbooks record that it marks bilateral contrast in its oldest function, that it applies systematically to local particles across the family, and that the gradational comparative is a later innovation in Greek and Indo-Iranian. What they do not explain is where the bilateral frame came from, or what the suffix was before it was a contrastive morpheme.

This paper proposes the missing etymology. Pre-PIE **der-* — the body-part word for "hand, open palm" that Paper 1 (Chavis & Claude 2026a) reconstructed on the basis of Greek δῶρον, Hesychian δάρις, Old Irish *dorn*, Welsh *dwrn*, and Latvian *dūre* — grammaticalized through HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE and became the suffix the handbooks reconstruct as **-tero-*. The spatial-direction function preserved across the family is what remains of the hand-noun's body-part semantics after the grammaticalization ran its course. Paper 3 (Chavis & Claude 2026c) completes the compositional analysis by showing that **der-* itself decomposes into bare **de-* (the hand-element, attested as the directional particle across the family) plus locative **-r-*, and that the same bare **de-* underlies the give-verb **deh₃-*. Paper 4 (Chavis & Claude 2026d) carries the framework into Anatolian, deriving Hittite *dā-* "to take" from the directive member of the locative-directive paradigmatic pair that Bauhaus (2019) has independently reconstructed for early PIE.

On the *t/d* alternation. The root is **der-* with original *d-*, following the consistent *d-* vocalism of the lexical cognates. The *t-* of the grammaticalized suffix reflects regular suffix-initial devoicing of *d-* in the PIE morphological system: PIE bound suffixes show a strong preference for voiceless obstruents in initial position (**-to-*, **-ter-/tor-*, **-tro-*, **-ti-*, **-tu-* are all voiceless-initial). When **de-* grammaticalized into a bound suffix, its initial *d-* devoiced to *t-* in conformity with this pattern.

2. The spatial-direction paradigm

We begin with the empirical center. Table 1 sets out the **-tero-* paradigm across the IE branches: every major local particle, the **-tero-* derivative it forms, and the spatial-direction adjective that results. Entries are restricted to inherited PIE-level formations.

PIE particle	Branch	*-tero- form	Derived meaning
* <i>ki-</i> "this, here"	Latin	<i>citer, citra</i>	"nearer, to this side"
* <i>ol-</i> / <i>uls-</i> "that, beyond"	Latin	<i>ulterior, ultra</i>	"farther, beyond"
* <i>ud-</i> "up"	Sanskrit	<i>uttara-</i>	"upper, northern"
* <i>ud-</i> / <i>uds-</i> "up, away"	Greek	ὑστερος (<i>hústeros</i>)	"later, after"
* <i>ádh-</i> / <i>ṇdh-</i> "below"	Sanskrit	<i>ádharma-</i>	"lower"
* <i>ádh-</i> "below"	Avestan	<i>aðara-</i>	"lower"
* <i>ṇdh-</i> "below"	Latin	<i>inferus</i> (bare alternant)	"lower"
* <i>(s)up-</i> / <i>uper-</i> "above"	Latin	<i>superus, super</i> (bare)	"upper, above"
* <i>(s)up-</i> "above"	Avestan	<i>upara-, upataro-</i>	"upper, on top of"
* <i>kṃt-</i> "below/alongside"	Hittite	<i>kattera-</i>	"lower, inferior"
* <i>h₁en-</i> "in"	Latin	<i>inter, intra</i>	"between, within"
* <i>h₁en-</i> "in"	Sanskrit	<i>ántara-</i>	"inner, between"
* <i>h₁en-</i> "in"	Greek	έντερα (<i>éntera</i> , n.pl.)	"innards" (lexicalized)
* <i>eks-</i> "out"	Latin	<i>exter, extra</i>	"outer, outside"
* <i>kom-</i> "with"	Latin	<i>contra</i>	"opposite, against"
* <i>post-</i> "after"	Latin	<i>posterus</i>	"later, behind"
* <i>pro-</i> "forward"	Av./OP	<i>frataro-</i>	"forward, more forward"
* <i>pro-</i> "forward"	Greek	πρότερος (<i>próteros</i>)	"former, before"
* <i>ápa-</i> "away"	Sanskrit	<i>ápara-</i>	"posterior, the other"
* <i>deks-</i> "right"	Latin	<i>dexter</i>	"right-handed"
* <i>deks-</i> "right"	Greek	δεξιτερός (<i>deksiterós</i>)	"right (hand)"
* <i>k^wo-</i> "who"	Latin	<i>uter</i>	"which of two"
* <i>k^wo-</i> "who"	Sanskrit	<i>katára-</i>	"which of two"
* <i>k^wo-</i> "who"	Greek	πότερος (<i>póteros</i>)	"which of two"
* <i>al-</i> "other"	Latin	<i>alter</i>	"the other (of two)"
(deictic) "other"	Greek	έτερος (<i>héteros</i>)	"the other"

Table 1. The PIE *-tero- spatial-direction paradigm across branches. Entries are restricted to inherited PIE-level formations. Bare alternants in -ero- are noted where attested.

The paradigm in brief. Latin preserves the most extensive paradigm. The bare alternant forms in *-trā* and *-trō* give the lexicalized adverbs *citra*, *ultra*, *intra*, *extra*, *contra*. Sanskrit preserves a closed paradigm of place-directional adverbs in *-tra* (the bare alternant): *átra* "here," *tátra* "there," *yátra* "where." The lexicalized neuter *āntrá-* "entrails" is worth noting: Latin *intestina*, Greek *έντερα*, and Sanskrit *āntrá-* all independently lexicalize the neuter of **h₁en-tero-* as "the inner parts" — three branches converging on the same lexicalization confirm the PIE-level age of the spatial adjective and its body-part orientation.

Avestan and Old Persian preserve the most archaic system. Milizia (2020: 268, 271) observes that the only Old Persian documented *-tara-/tama-* formations are derivatives of local particles or preverbs. There are no gradational comparatives in *-tara-* in Old Persian — reflecting, on Milizia's reading, the most archaic preserved layer of the formation. Hittite preserves only one **-tero-/*-(e)ro-* form productively: *kattera-* "lower; inferior; further along" (Kloekhorst 2008: 538–539), built on the "below"-particle *katt-*. This form is the empirical anchor of §6.

The Handbook's verdict. The *Handbook of Comparative and Historical Indo-European Linguistics* (vol. 3) is explicit: outside the Greek and Indo-Iranian extension into gradational comparison, **-tero-* and **-(e)ro-* "are confined to derivations of pronouns and particles." The original PIE function of the suffix is exactly the paradigm documented in Table 1. The framework's contribution is an etymological account of why that particular suffix should perform that particular function — something the standard treatment leaves open.

3. The bilateral character of **-tero-* and the ambidexter argument

The standard handbooks agree that **-tero-*'s original function was bilateral contrast — not general gradation, not loose comparison, but specifically "one of a pair as opposed to the other member of the pair." Fortson's formulation is precise: the basic function was to "set off a member of a pair contrastively from the other member of the pair, as in Greek *skaiós* 'left' vs. *deksiterós* 'right,' i.e. 'the one on the right (as opposed to the one on the left)'" (Fortson 2010). Sihler (1995: 360–361) gives the same analysis. The canonical examples — *póteros* "which of two" and *deksiterós* "right (as opposed to left)" — confirm it. The bilateral character is the mainstream consensus. What the mainstream does not provide is any account of where that bilateral character comes from.

The framework answers that question. Bare **de-* — the directional particle documented in Paper 3 as the morpheme underlying the Greek allative *-δε*, Old Church Slavonic *do*, and English *to* — is an absolute directional: "toward there," with no implied alternative and no bilateral frame. Greek *oíkade* "homeward" does not presuppose a contrasted alternative. The locative **-r-* element changes this. It anchors the directional to the body as frame of reference, and bodies have inherent bilateral symmetry: exactly two sides, symmetrically opposed. The absolute directional becomes a relational one. The bilateral character was established in the suffix's earliest productive use — with **deks-* "right," giving **deks-i-tero-* "right as opposed to left," where the two hands provided the bilateral frame.

Ambidexter is the compound that makes this argument explicit. It parses as *ambi-* (PIE **h₂mb^hi* "on both sides") + **deks-* ("right") + *-i-* (linking vowel) + **-tero-* + Latin nominal ending: "having the right-side marker on both sides." That statement is only meaningful if **-tero-* normally marks one of two bilateral sides.

You can only say "on both sides" if the default is "on one side." The compound's structure presupposes and thereby reveals the bilateral character the framework derives from the hand-noun. Festus's gloss confirms the parse: *ambidexter qui dexteris ambo manus habet*. The Greek cognate *deksiterós*, inherited independently from PIE, confirms that the bilateral frame is not a Latin formation but PIE.

4. The directionality question

4.1 The standard view and current developments

The standard view, summarized by Szemerényi (1996: 197), holds that the bare suffix **(e)ro-* is original and the *t*-bearing **-tero-* is a secondary extension. This is the view of Wackernagel & Debrunner (1957) for Sanskrit, Schwyzler (1939: 533) for Greek, and Sihler (1995: 360–361) and Weiss (2009: 357–358) for the comparative treatment generally.

Two recent treatments represent the most sophisticated elaborations of the metanalysis account. Bauhaus (2019) argues that bare **-r* was a true PIE locative case ending (not an "Adverbialendung") whose reflexes survive in pronominal and nominal formations across the family, and that the **(t)ero-* variants arose by "false segmentation of stems ending in *-t*": forms like **k_mt-er* "at the underside" were reanalyzed by speakers as stem + new suffix **-ter*, which then spread productively (Bauhaus 2019: §4). Kahl (2024), in a Harvard dissertation on Indo-European degree morphology, independently argues for a locative origin of **(e)ro-* and derives **(t)er/*-tero-* by metanalysis from *t*-final stems. Kahl's §2.3.1 states explicitly: "the origin of the 'secondary comparative' of Indo-Iranian and Greek is traced to an archaic locative morpheme **-er*."

4.2 Where the framework agrees with Bauhaus and Kahl

On the locative status of **-r/-er-*, the framework and both Bauhaus and Kahl are in complete agreement. Bauhaus: "we are dealing with a real case ending, not an 'Adverbialendung.'" Kahl: the suffix is an "archaic locative morpheme." The framework: the **-r-* of **der-* and **-tero-* is the locative morpheme documented in PIE heteroclitic neuters (Lipp 2009), nominal formations (Bauhaus 2019), and degree morphology (Kahl 2024). Paper 3's §3 cites all three treatments as independent published convergence on the locative **-r-*. That convergence is the morphological foundation on which the framework builds, and it now rests on three independent scholarly traditions.

On the broader morphological landscape, the framework and Bauhaus's account also agree: nominal roots take locative **-er* to give locative formations, which thematize to give **-ero-/*-tero-* contrastive adjectives. Bauhaus's published inventory of locatives — **h₁up-er* "on top," **h₂ep-er* "in the back," **k(e)nt-er* "at the underside," **ndh-er* "at the bottom" — gives the framework's **de-r-* "at the

place of the hand-direction" exactly the morphological class it belongs to. The body-part noun is a member of Bauhaus's documented locative-formation paradigm.

Additionally, Bauhaus (2019: §5) reconstructs a paradigmatic opposition between locative **-r* and directive **-o* on the same nominal roots — **h₁upér* / **h₁upó*, **k(e)nter* / **knto*, **ndher* / **ndho* and related pairs — noting that the directive **-o* "could keep its productivity at least until Anatolian" while the locative **-r* "ceased to be productive in all attested languages." This paradigmatic pair provides published external precedent for the structural opposition Paper 4 reconstructs for the hand-root: locative **de-r-* (the body-part noun) versus directive **de + a* (Hittite *dā-* "to take"). The framework applies Bauhaus's documented structural type to a new lexical item.

4.3 Where the framework disagrees with Bauhaus and Kahl

The disagreement is specific: the directionality of the /t/. Bauhaus and Kahl hold that bare **(e)ro-* is original and **-tero-* arose by metanalysis. The framework holds the reversal: bare **(e)ro-* is secondary, arising by OCP-driven suffix-initial deletion before stop-final bases, and **-tero-* is original — with the /t/ the initial consonant of the hand-noun **de-*, devoiced in suffix-initial position. The distributional data (§7) are compatible with both accounts; the two hypotheses are distinguished by their etymological commitments, not by the data alone.

What is striking, and not accounted for by Bauhaus's own framing, is the composition of his locative-formation inventory. Every bare **-er* form he reconstructs — **h₁up-er* (labial stop), **h₂ep-er* (labial stop), **k(e)nt-er* (dental stop), **ndh-er* (voiced aspirate dental), **dhg̃zm-er* (dental + cluster), **h₂us-er* (sibilant), **nokwt-or* (dental stop), **ghim-er* (cluster) — lands on a base ending in a stop or stop-cluster. This is precisely the phonological environment where the framework's OCP-driven deletion rule predicts the suffix-initial /t/ should drop. Under Bauhaus's account this is the pool from which metanalysis drew; under the framework's account it is the expected output of a phonological rule. Both analyses accommodate the data. The framework's account is more constrained because it derives the distribution from a single rule.

4.4 What would distinguish the hypotheses

The framework's reversal predicts that bare **(e)ro-* forms arise by OCP-driven cluster simplification: suffix-initial /t/ of **-tero-* is deleted when the base ends in a stop, creating a stop+stop cluster that violates the OCP. Under metanalysis, the same distribution follows from the source-form environments (stems ending in /t/ would naturally not include stop-final bases). Both hypotheses therefore predict approximately the same surface distribution. What the distributional test of §7 does decide against is the older, looser version of the standard view — in which the *t*-extension is treated as analogically irregular, with no systematic

phonological conditioning. The Szemerényi/Sihler framing leaves the distribution as a lexical accident. The data argue against that framing regardless of which account of the /t/ direction is correct.

5. The reversal: existing phonological rules deliver the bare form

The phonological mechanism by which an original **-tero-* could yield bare **(e)ro-* in some environments does not require a new sound rule. The mechanism is cluster simplification at morpheme boundaries, driven by the Obligatory Contour Principle (OCP), as documented for PIE by Byrd (2015).

Byrd's framework is the current authoritative treatment of PIE syllable structure. The OCP — the principle that PIE banned geminates across morpheme boundaries — is documented as undominated in PIE following Meillet (1934). When morpheme-boundary clusters violated OCP, PIE had two resolution strategies: s-epenthesis for VTTV sequences (**úeid-tó- > *úeit.stó-*), and consonant deletion for VTTRV sequences where epenthesis would create an illegal coda (**méd-trom > Greek métron*). The latter is the Métron Rule (Saussure 1885; Mayrhofer 1986: 111; Byrd 2015: ch. 3.6).

The Métron Rule is exactly the mechanism needed. It deletes a dental in dental-dental-resonant sequences at morpheme boundaries. Crucially, Byrd is explicit that the rule's deletion target is genuinely undecided: his footnote 161 reads: "If the /d/ of the root was lost, we would expect *mé.trom*; if the /t/ of the suffix was lost, we would expect *mét.rom* (see chapter 7 for discussion)." Greek *métron* with single -t- is consistent with both outcomes. The Métron Rule therefore has two alternative outcomes; the specific case does not disambiguate. We propose that the same rule applies here: pre-PIE **X-tero-* (where X is a base ending in a dental, as in **k̑mt-tero-*) is reduced to **X-(e)ro-* by deletion of the suffix-initial dental. The rule is not new. The deletion target is the alternative Byrd himself explicitly recognizes.

6. The Hittite test case: kattera-

The Hittite contrastive adjective *kattera-* "lower; inferior; infernal; further along" (Kloekhorst 2008: 538–539) is the empirical anchor for the phonological argument. It is built on the "below"-particle **k̑mt-* that gives Latin *cum*, Old Irish *cét*, Greek *katá*, Hittite *katta*.

The derivation under the framework runs in four steps: pre-PIE **k̑mt-tero-* (root + suffix; dental cluster across morpheme boundary) → PIE **k̑mt-ero-* (OCP / Métron Rule; suffix-initial *t* deleted) → Anatolian **katt-era-* (**m̑* vocalizes to *a*; **m* before **t* assimilates to geminate) → Hittite *kattera-* (regular Anatolian outcome, Melchert 2017: 11–12).

The alternative deletion target — root-final *t* — would have predicted Hittite *katera-* with single intervocalic *-t-*, not the attested geminate *-tt-* of *kattera-*. The Hittite outcome picks the suffix-initial deletion. This is the empirical bite of Byrd’s footnote 161: there are surface forms that disambiguate the rule’s outcome, and Hittite *kattera-* is one of them. Both the framework and the metanalysis account predict the same surface form, but the framework derives the Hittite geminate from a principled phonological rule rather than from an analogical process.

Lipp’s (2017) semantic analysis of **k̑mt-* is worth noting. In his GeSuS abstract on Hittite *katta*, Lipp glosses **k̑mt-* not merely as “below” but more precisely as “into close contact, into/under pressure” — a reading that bridges the apparently divergent senses “alongside” (Latin *cum*, Greek *katá* in some uses) and “downwards” (Hittite *katta*, *kattera-*) under a single spatial concept. Under Lipp’s gloss, the contrastive **k̑mt-tero-* “the alongside-or-downward-side one (of two)” parses with greater semantic transparency than the simpler “below”-gloss alone permits.

7. The distributional test of the reversal hypothesis

7.1 The prediction

Under the reversal hypothesis, bare **(e)ro-* forms arise by OCP-driven cluster simplification: the suffix-initial */t/* of **-tero-* is deleted when the base ends in a stop. The reversal hypothesis therefore predicts that bare **(e)ro-* forms should cluster systematically on stop-final bases, while *t*-bearing **-tero-* forms should cluster on bases ending in vowels, sonorants, or fricatives. Under metanalysis (Bauhaus 2019; Kahl 2024), the same distribution is predicted because the metanalysis source forms were *t*-final stems, which would naturally not include further stop-final bases. Both hypotheses predict the same distribution and are not distinguished by the test; what the test distinguishes both from is the older “irregular extension” framing.

Branch	Bare <i>*(e)ro-</i> on stop-final bases	<i>t</i> -bearing <i>*-tero-</i> on non-stop bases
Sanskrit	<i>ádharma-</i> (<i>*ṇdh-</i> , dental asp.) <i>ápara-</i> (<i>*ap-</i> , labial stop) <i>ávāra-</i> (<i>*au-</i> , glide)†	<i>uttara-</i> (<i>*ud-</i> , dental → gemination)
Avestan	<i>upara-</i> (<i>*up-</i> , labial stop) <i>aḍara-</i> (<i>*adh-</i> , dental) <i>apara-</i> (<i>*ap-</i> , labial stop)	<i>fratarā-</i> (<i>*pro-</i> , vowel)
Latin	<i>super</i> (<i>*(s)up-</i> , labial stop) <i>inferus</i> (<i>*ṇdh-</i> , dental)	<i>inter</i> , <i>intra</i> (<i>*en-</i> , nasal) <i>exter</i> , <i>extra</i> (<i>*eks-</i> , fricative) <i>ultra</i> (<i>*ol-</i> , liquid) <i>citra</i> (<i>*ki-</i> , vowel) <i>contra</i> (<i>*kom-</i> , nasal)
Greek	ὐπέρ (<i>*(s)up-</i> , labial stop)	πρότερος (<i>*pro-</i> , vowel) ἑντέρα (<i>*en-</i> , nasal) δεξιτερός (<i>*dek̑si-</i> , vowel)

† *ávāra-* (**au-*, glide-final) is the weakest match: the base ends in a glide rather than a stop. We treat it as a lexicalized relic; it does not falsify the prediction.

7.2 Results and exceptions

Latin shows the most extensive paradigm: seven of seven inherited forms match the prediction. Two Latin forms do not fit: *subter* "below" (base **(s)up-*, labial stop) and *propter* "near" (base **prop-*, labial stop). Both are Latin-internal formations modeled on existing **-tero-* words, as Lewis & Short and de Vaan (2008) treat them. They illustrate what unconstrained analogical extension looks like once the original conditioning rule has ceased to be phonologically active: Latin adds *-ter-* freely to stop-final bases because the OCP constraint that conditioned the original alternation no longer applies. This is itself evidence for the reversal hypothesis: the inherited paradigm shows systematic conditioning, while the later Latin extensions do not.

The pronominal possessive paradigm is the test's failure mode, acknowledged openly. First- and second-plural possessives show the alternation breaking down across branches: Latin *noster*, *vester* have **-tero-* on sibilant-final bases; Gothic *unsar*, *izwar* have bare **(e)ro-* on similar bases. This is not predicted by any phonological rule. The defense is that the pronominal possessives are a small functional category that became generalized branch-internally once the original conditioning rule had ceased to operate — Latin generalizing **-tero-*, Germanic the bare **(e)ro-*.

7.3 What the test shows and what it does not

Run on the inherited PIE positional adjective paradigm, the prediction is matched in approximately 19 of 19 etymologically transparent cases across Sanskrit, Avestan, Latin, and Greek. The pattern is not analogically irregular. The inherited paradigm shows systematic phonological distribution.

What the test does not show is which of the two competing accounts is correct. The framework's reversal predicts the systematic distribution from the OCP; Bauhaus's and Kahl's metanalysis predicts the same distribution from the source-form environments. Both are compatible with the data. The contribution of the test is therefore narrower than earlier drafts of this paper claimed: it argues against the "irregular extension" framing of Szemerényi, Sihler, Wackernagel & Debrunner, and Schwyzler, but does not by itself decide between the framework and the metanalysis accounts. The framework's advantage over metanalysis is etymological: the framework explains why the *t-* was there in the first place, as the initial consonant of the hand-noun **der-* devoiced in suffix-initial position. Metanalysis leaves the source of the *t-* open.

8. The Anatolian distribution argument

Of all the distributional facts in the IE contrastive system, the Anatolian asymmetry is the most informative. Anatolian preserves the bare **(e)ro-* in a lexicalized formation (*kattera-*) but shows no productive **-tero-* whatever. The productive contrastive function is taken over by **-eti-Ho-* (Hittite *-ezziya-*), in

ḥantezziya- "first, foremost" and *appezziya*- "back, last." The standard view treats this as an arbitrary inheritance: Anatolian "happens" to have generalized one alternant. The reversal converts this coincidence into a structural prediction.

The reasoning is as follows. If the *t*-bearing **-tero-* is original at the pre-PIE stage, and the bare **(e)ro-* arises by cluster simplification in lexicalized formations, then at any given stage of the proto-language we expect: productive **-tero-* derivations on stems where no cluster simplification applies, and lexicalized **(e)ro-* relics in formations where the simplification has applied. The productive system is always **-tero-*; the bare alternants are always relics.

Now consider the Anatolian split. If Anatolian split off from PIE early — as the Indo-Anatolian hypothesis maintains, with positive evidence from morphology, vocabulary, and verb structure (Kloekhorst 2008: 7–11; Melchert 2014; Kloekhorst & Pronk 2019) — then it represents a stage of the proto-language earlier than PNIE. At that stage, the productive **-tero-* system may not yet have generalized across the morphology. What had stabilized were the lexicalized formations on the most frequently-occurring local particles — like **ḱmt-* "below" — where the simplification had already produced bare forms. These bare forms, lexicalized as adjective stems, were available for inheritance into daughter branches.

Under this picture, Anatolian inherits exactly what it shows: bare relics in lexicalized contrastive formations (*kattera-*) but no productive **-tero-* system. The productive system develops post-Anatolian, in PNIE, when the suffix generalizes across the contrastive paradigm in non-cluster environments. Sanskrit, Greek, Italic, Tocharian, and Germanic all inherit the productive system because they descend from the post-Anatolian PNIE. Anatolian does not, because its split predates the generalization. Under the standard view, the Anatolian distribution is unmotivated. The framework converts it from a coincidence into a prediction.

This Anatolian-distribution argument converges with the independent evidence of Paper 4 (Chavis & Claude 2026d). Bauhaus (2019: §5) independently observes that the directive **-o* "could keep its productivity at least until Anatolian" while the locative **-r* "ceased to be productive in all attested languages" — which is precisely the asymmetry the framework predicts. The locative member of the paradigmatic pair fossilized into body-part lexical items; the directive member remained productive through the Anatolian stage and gave Hittite *dā-* "to take."

9. Cross-cultural evidence: HAND > BINARY-CONTRASTIVE

The framework's morphological claim has a typological prediction: similar pathways or component links should be attested in other languages. The direct pathway HAND > BINARY-CONTRASTIVE is not catalogued in Heine & Kuteva (2002), the standard reference on grammaticalization pathways. What is documented are HAND > AGENT MARKER, HAND >

LOCATIVE/GENITIVE/POSSESSIVE, HAND > FIVE, and HAND > MANNER. The closest is HAND > LOCATIVE, which forms one of the two component links needed.

HAND > SIDE is documented in P'urhepecha (the suffix *-k'u* from *jak'i* "hand" marks the locative on body parts), Mordvin (*ked'* "hand" shows partial application of the body-part-to-spatial pathway), and the Mayan languages (Kaqchikel, Tz'utujil, K'iche', Tzotzil use body-part terms productively as relational nouns expressing spatial locations). SPATIAL > BINARY-CONTRASTIVE is documented in Tungusic and Uralic.

Kahl (2024) documents typological parallels for binary-contrastive morphemes that bear directly on the framework. In Uralic, Proto-Uralic **-mpV* is reflected in Finnish as the productive comparative but preserves "synchronically unusual binary-contrastive forms" — the relative pronoun *jompi* and its interrogative equivalent *kumpi* "which of two" — exactly the functional pairing the reversal hypothesis predicts for PIE **-tero-*. In Tungusic, the suffix Proto-Tungusic **-dimar* appears both as an optional comparative marker and as a binary-contrastive marker. Kahl notes that "it seems that the binary-contrastive function is older and was secondarily drawn into comparative constructions" — precisely the diachronic ordering the framework proposes for PIE: binary-contrastive **-tero-* generalized to gradational comparison in the Greek and Indo-Iranian innovation.

There is one piece of cross-linguistic evidence for HAND used directly for binary contrast that requires no typological reconstruction: the English idiom "on the one hand...on the other hand." When a speaker says "on the one hand the evidence supports X; on the other hand it supports Y," HAND is doing the work of marking the two terms of a binary opposition. The construction is alive, productive, and unmistakable. It is not in Heine & Kuteva's database, presumably because it is idiomatic rather than fully grammaticalized into a bound morpheme. But the function it performs is exactly the function claimed for pre-PIE **der-* in its grammaticalized form.

The argument rests on nine distinct lines of evidence, collected in Table 2.

#	Reason	Evidence
1	Lateralization domain is hand-related	Suffix productively forms hand-side terms: Latin <i>dexter</i> , Greek δεξιτερός, OE <i>swiþro</i> "right (hand)" (§3).
2	Body parts lexicalize from the spatial adjective	The neuter of <i>*h₁en-tero-</i> "inner" lexicalizes as "entrails" independently in three branches: Latin <i>intestina</i> , Greek έντερα, Sanskrit <i>āntrá-</i> (§2).
3	Paradigm matches grammaticalization output	Every PIE local particle takes <i>*-tero-</i> ; result names the spatial contrastive (Table 1). Exactly the output HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE predicts.

#	Reason	Evidence
4	Binary-contrastive uses a hand-pivot in living language	*-tero- marks "which of two" (Latin <i>uter</i> , Sanskrit <i>katára-</i> , Greek <i>πότερος</i>). English "on the one hand...on the other hand" uses HAND as explicit binary-contrast pivot (§9).
5	Component pathways independently attested	HAND > SIDE in P'urhepecha, Mordvin, Mayan (Heine & Kuteva 2002). SPATIAL > BINARY-CONTRASTIVE in Tungusic, Uralic, Saami (Kahl 2024).
6	Diachronic ordering independently attested	Tungusic shows BINARY-CONTRASTIVE > COMPARATIVE — exactly the ordering PIE shows, with gradational comparison as the late Greek/Indo-Iranian innovation (Kahl 2024).
7	<i>Ambidexter</i> presupposes the bilateral norm	Latin <i>ambidexter</i> = <i>ambi-</i> + <i>*deks-</i> + <i>-i-</i> + <i>*-tero-</i> : "on-both-sides + right-handed + CONTRASTIVE." Only meaningful if <i>*-tero-</i> normally marks one of two bilateral sides (§3).
8	Anatolian distribution becomes a prediction	Anatolian preserves a lexicalized bare relic (<i>kattera-</i>) but no productive <i>*-tero-</i> . Under the framework, this follows from the chronology of the Anatolian split. Under the standard view, it is unmotivated (§8).
9	Bauhaus (2019) independently supports the locative <i>*-r-</i> and the paradigmatic pair	Bauhaus reconstructs <i>*-r</i> as a true PIE locative case ending and the locative/directive pair (<i>*h₁upér</i> / <i>*h₁upó</i> etc.) as a documented PIE structural type. The framework's <i>*de-r-</i> (locative) / <i>*de + a</i> (directive, Paper 4) fits this published class directly (§4.2).

Table 2. Nine reasons **-tero-* is semantically consistent with an original hand meaning. No single line forces the framework; the convergence is what is claimed.

10. Future research

This paper has restricted itself to the contrastive **-tero-*. Three morphological extensions that the framework also touches will each require their own treatment.

The kinship suffix **(h₂)ter-* — in **māter-* "mother," **ph₂ter-* "father," **b^hrāter-* "brother," **d^hugh₂ter-* "daughter" — shares the *t-r* structure of the contrastive **-tero-*. Whether it descends from the same pre-PIE etymon, perhaps via a possessive/relational pathway (HAND > GENITIVE/POSSESSIVE in Heine & Kuteva 2002) and a kinship-classifier development, is an open question. Latin *matertera* "maternal aunt" — a Latin-internal extension of **-tero-* to a kinship term — hints at the pathway.

The agent suffix **-ter-/tor-* — in Greek *dó-tor*, Latin *da-tor*, Sanskrit *dā-tār-* — similarly shares the *t-r* structure. The cross-linguistic pathway HAND > AGENT MARKER is among the well-documented HAND grammaticalizations (Heine & Kuteva 2002: 167–168). Whether the PIE agent suffix descends from the same pre-PIE etymon is the natural extension of the present paper.

The broader hand-semantics network — including **dher-* "hold, maintain" with its reflex in Sanskrit *dharma*, and the directional verb **terh₂-* "to cross over" — is mapped in the future-research section of Paper 3 (§7). Paper 4's derivation of Hittite *dā-* from pre-PIE **de + a* completes the four-paper framework as published; the extensions noted above represent the next research horizon.

11. Remaining problems

Three problems deserve flagging, none of which we regard as fatal.

First, the directionality question is genuinely contested. The standard view (Bauhaus 2019; Kahl 2024) and the framework's reversal are both consistent with the distributional data. The framework provides independent reasons for the reversal — the etymological motivation of the hand-noun source, the Hittite *kattera-* anchor, and Byrd's explicitly recognized alternative deletion target — but the distributional test of §7 does not by itself decide between reversal and metanalysis.

Second, the proposed grammaticalization pathway from HAND to the contrastive function is supported by typological parallels in the component links but does not have direct PIE-internal documentation. The *deksiterós/dexter* compounds preserve the lateralization function transparently, but the further step — from lateralized relational form to general spatial-direction marker on local particles — is reconstructed rather than directly attested.

Third, the timing of the various grammaticalizations relative to PIE-internal sound changes is not fully worked out. The cluster simplification is placed at pre-PIE/early PIE, before the Anatolian split, but a fuller chronology remains to be developed.

12. Conclusion

The morphological argument rests on a paradigm-level observation: the PIE contrastive suffix **-tero-* is, in its oldest documented function, the systematic morpheme for spatial-direction marking across the Indo-European family. Every major PIE local particle has a **-tero-* derivative naming the spatial-contrastive of that particle, from Latin *citra/ultra/intra/extra/contra* to Sanskrit *ádharma-/uttara-/ápara-/ántara-* to Avestan and Old Persian *fratara-/apatara-/upatara-* to Hittite *kattera-*. The Handbook states explicitly that this spatial-particle function is the original PIE function; the gradational "more X" use is a Greek and Indo-Iranian innovation.

The framework of Paper 1 — pre-PIE **der-* "hand, open palm" — provides exactly the etymological motivation the standard treatment lacks. A hand-noun grammaticalizing through HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE is exactly what one would expect to find as the systematic spatial-direction marker on local particles. The mainstream consensus that **-tero-* marks bilateral contrast is not the framework's claim; it is the established starting point. The framework answers what the mainstream leaves open: where the bilateral character came from.

The Latin compound *ambidexter* presupposes the bilateral norm by negating it; Hittite *kattera-* anchors the OCP phonology; the distributional test argues against

the "irregular extension" framing; the Anatolian distribution — bare relic but no productive **-tero-* — is a structural prediction of the framework, unmotivated on the standard view; and Bauhaus's (2019) independently published locative-directive paradigmatic pairs provide external precedent for the structural opposition the framework reconstructs across Papers 3 and 4. Nine convergent reasons are collected in Table 2.

We do not claim the hypothesis is correct in all its details. The directionality is contested; the grammaticalization pathway is reconstructed rather than directly attested; the pre-PIE chronology is open. We do claim that the hypothesis is internally consistent, provides a unified explanation where the standard treatment provides only description, and deserves examination as an alternative reading of a morphological landscape that has been catalogued but not etymologized for over a century.

13. Probability estimate

Following the practice established in Paper 1, we close with an explicit probability assessment framed conditionally.

Conditional on the central claim of Paper 1 being correct — that pre-PIE had a body-part word **der-* meaning "hand, open palm" — our estimate of the probability that the substantive thesis of this paper is correct in its basic outline is approximately **35-45%**. This is a modest upward revision from v7's range, reflecting the strengthening of the locative **-r-* claim with Bauhaus (2019) and Kahl (2024) as two further independent published treatments, and the recognition that Bauhaus's own locative-formation inventory is consistent with the OCP-distributional prediction.

The marginal probability — unconditioned, integrating over the probability that Paper 1 is also correct — is approximately the product: 25-35% (Paper 1 v26) × 35-45% (Paper 2 conditional on Paper 1) ≈ 8.75-15.75%. We continue to regard the central claim as more likely wrong than right at the marginal level, and to regard it as interesting enough, and the evidential structure dense enough, that publication for examination by other scholars is warranted.

14. Authorship and contributions

This is the eighth version of Paper 2 in the *der-* project. It builds on Paper 1 (Chavis & Claude 2026a) and is developed further in Papers 3 and 4 (Chavis & Claude 2026c, 2026d). We continue the same co-authorship arrangement.

Chavis's contributions. The substantive intellectual direction originates with Chavis. The hypothesis that **der-* "hand" underlies the contrastive **-tero-* was first articulated by Chavis, as was the slogan "the *ter* in **-tero-* is hand." Chavis identified *ambidexter* as the morphologically transparent witness and directed the

investigation toward the bilateral-contrast argument of §3. Chavis insisted that the phonological argument rely entirely on existing PIE rules rather than positing a new sound law — a constraint that led to the engagement with Byrd (2015) and the identification of footnote 161 as the relevant mechanism. The distributional test in §7, the Anatolian distribution argument in §8, the engagement with Kahl (2024), and the updated engagement with Bauhaus (2019) were all directed by Chavis.

Claude's contributions. Claude's contributions are primarily in source synthesis, exposition, and analytical development. Claude searched and read the relevant published literature and identified the key new contributions from Bauhaus (2019) and Kahl (2024) for the locative *-r- claim. Claude identified Byrd's footnote 161 as the explicit recognition of suffix-initial deletion; documented the Latin *-trā/-trō* paradigm; developed the distributional test cases; surfaced the typological parallels (Uralic, Saami, Tungusic) for §9; proposed the Anatolian distribution argument of §8; and identified the striking fact that Bauhaus's own bare-locative inventory clusters on stop-final bases consistent with the OCP prediction. The probability estimate in §13 is Claude's own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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